LiDAR Introduction and Use

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1. Course Content

- Understand the LiDAR's basic parameters.
- Start the example program and view the LiDAR's visualized point cloud.

2. LiDAR Introduction

- The ROSMASTER-M1Pro package provides two LiDARs: 1. Tmini Plus, 2. Slamtec C1

2.1 T-mini Plus

Attribute	Value	Unit
Ranging frequency	4000	Hz
Scan frequency	6 (6-12)	Hz
Ranging range	0.05-12	m
Scan angle	0-360	Deg
Ranging accuracy	20	mm
Angle resolution	0.54	Deg

2.2 Slamtec C1

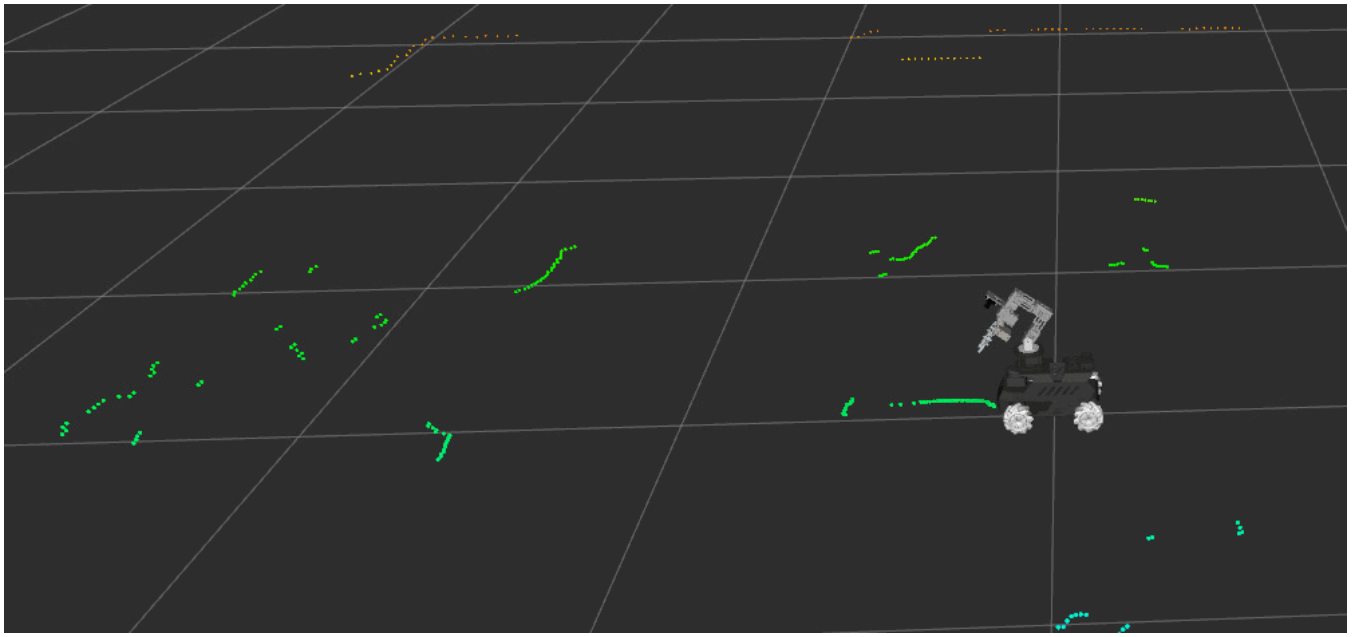
Attribute	Value	Unit
Ranging frequency	4000	Hz
Scan frequency	(6-12)	Hz
Ranging range	0.05-12	m
Scan angle	0-360	Deg
Ranging accuracy	20	mm

Attribute	Value	Unit
Angle resolution	0.54	Deg

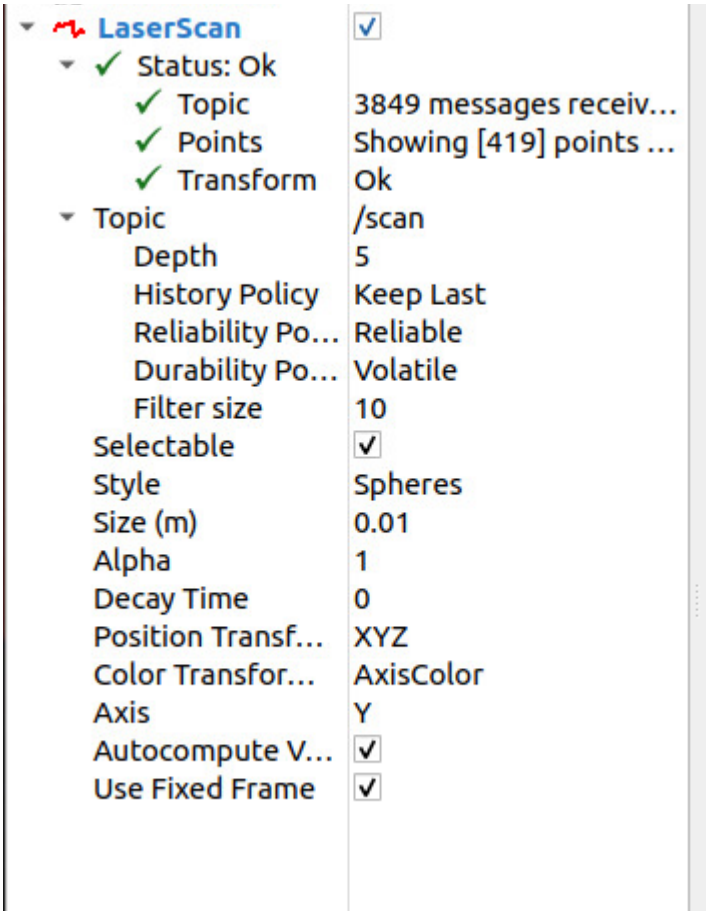
3. Case Demonstration

```
ros2 launch yahboomcar_bringup laser_view.launch.py
```

The green point cloud in the RVIZ image is the point cloud from the LiDAR scanning the contours of the surrounding environment



- In the **LaserScan** option in rviz's sidebar, you can view the received LiDAR data information and modify the LiDAR point cloud's color style, etc.



4. Viewing LiDAR Data

- Run the ros2 CLI command

```
ros2 topic echo /scan
```

```
jetson@yahboom:~$ ros2 topic echo /scan
header:
  stamp:
    sec: 1787820015
    nanosec: 400578297
  frame_id: laser
angle_min: -3.1415927410125732
angle_max: 3.1415927410125732
angle_increment: 0.0087387841194868 09
time_increment: 8.943522698245943e-05
scan_time: 0.0643039271235466
range_min: 0.05000000074505806
range_max: 16.0
ranges:
- 2.317500114440918
- 2.3459999561309814
- 2.3515000343322754
- 2.3515000343322754
- 2.349250078201294
- 2.3545000553131104
- 2.371000051498413
- 2.371000051498413
```

A single frame of data is as shown above, where:

- frame_id: represents the LiDAR's coordinate frame name.
- angle_min and angle_max: represent the minimum and maximum radians of the LiDAR scan; converted to degrees here, they are 0 degrees to 360 degrees.
- angle_increment: represents the angle increment.
- range_min and range_max: represent the minimum and maximum distance of the LiDAR scan, in meters
- ranges: represents the distance measured at each angle within the LiDAR's scan range, in meters.

5. Source Code Analysis

- car_base launch file path

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/launch/car_base.launch.py
```

- `GetLidarType()`: gets the current LiDAR model, and starts the corresponding driver respectively according to the return value

- When the model is `c1` (Slamtec C1): starts the `sllidar_node` node, configuring the serial port `/dev/rplidar`, baud rate `460800`, coordinate frame `frame_id` as `laser`, and enabling angle compensation `angle_compensate`
- When the model is `tmini` (T-mini Plus): calls `ydlidar_launch.py` via `ExecuteProcess` to start the driver
- Both are controlled by the `use_lidar` launch parameter; when it is `False`, the LiDAR is not started

```
def generate_launch_description():
    #bring up LiDAR sensor
    LiDAR_Type=GetLidarType()
    if LiDAR_Type=="c1":
        bringup_LiDAR =Node(
            package='sllidar_ros2',
            executable='sllidar_node',
            name='sllidar_node',
            parameters=[{'channel_type': 'serial',
            'serial_port': '/dev/rplidar',
            'serial_baudrate': 460800,
            'frame_id': 'laser',
            'inverted': False,
            'angle_compensate': True,
            'scan_mode': 'Standard',
            'masked_sector_center_deg': 0.0,
            'masked_sector_half_width_deg': 57.0}],
            output='screen',
            condition=IfCondition(LaunchConfiguration('use_lidar')),
            arguments=['--ros-args', '--log-level', 'warn'])
    elif LiDAR_Type=="tmini":
        bringup_LiDAR = ExecuteProcess(
            cmd=['ros2', 'launch', 'ydlidar_ros2_driver', 'ydlidar_launch.py'],
            output='screen',
            condition=IfCondition(LaunchConfiguration('use_lidar')))
```

- laser_view launch file path

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/launch/laser_view.launch.py
```

- Loads the Rviz configuration file `laser_view.rviz`, in which the LiDAR point cloud display style is preset
- Includes `car_base.launch.py` via `IncludeLaunchDescription`, enabling only the LiDAR (`use_lidar: True`) and disabling the IMU, camera, magnetometer, etc.

```
def generate_launch_description():

    declared_arguments = []
    declared_arguments.append(
        DeclareLaunchArgument(
```

```
            "rviz",
            default_value="true",
            description="Start Rviz2 automatically",
        )
    )

    rviz_config_file= os.path.join(get_package_share_directory('yahboomcar_bringup'),
'rviz', 'laser_view.rviz')
    bringup_laser_launch=IncludeLaunchDescription(

 PythonLaunchDescriptionSource(os.path.join(get_package_share_directory('yahboomcar_bringup'
),'launch','car_base.launch.py')),
            launch_arguments={'log_level': 'info',
                              'use_imu': 'False',
                              'use_lidar': 'True',
                              'use_mag': 'False',
                              'use_camera': 'False',
                              'odom_tf_source': 'base'
                              }.items())
    rviz_node=  Node(
        package='rviz2',
        executable='rviz2',
        name='rviz2',
        arguments=['-d', rviz_config_file],
        output='screen',
        condition=IfCondition(LaunchConfiguration("rviz"))
    )
```